

Shenandoah Valley Mobile Beef Harvest & Processing

Business Plan – Research Draft

October 2, 2026

Read this first. This is a research-stage draft assembled from 71 public sources. It is not legal, tax or financial advice. Every figure is dated to its source, and the financial projections are **illustrative**: they combine sourced benchmarks with clearly labelled placeholder assumptions that must be replaced with real quotes, a zoning determination and lender terms before the plan is used to raise money. Items marked *[inferred]* are this plan's reasoning, not a source's statement.

1. Executive Summary

The business. A custom-exempt mobile beef service for Virginia's northern Shenandoah Valley. A mobile unit **harvests cattle on the owner's farm**. The carcass hangs and ages for 7–21 days in a **company-owned cooler**. The same mobile unit then **cuts, vacuum-packs and labels** the beef while docked at the cooler. Customers are cattle owners (freezer-beef producers) and the households that buy their animals in halves and quarters before slaughter.

Why now.

- Regional USDA plants ran at **125–150% of normal capacity** in 2021, with producer waits of 12–18 months.
- Virginia's own five-year plan records **18–24 month waits**, explicitly backs investment in custom-exempt capacity, and notes that the whole state depends on **a single renderer**.
- The nearest USDA plant to Winchester still books **3–5 months out** in 2026.

Why this model.

- **It's legal.** VDACS expressly allows a custom-exempt “mobile slaughter and processing unit” under a **free annual Custom Permit**. There is no inspector to schedule, no HACCP validation, and no per-site federal paperwork.
- **It avoids the usual failure.** The documented reason mobile units fail is the lack of a partner to hang and cut their carcasses: one California unit sat idle for about two years. Owning the cooler and the cutting removes that dependency.

What it is not. Custom-exempt beef is marked “Not for Sale”, so it **cannot be sold at retail, to restaurants, at farmers' markets or online**. Customers must own the live animal before slaughter. Moving to Virginia state inspection or USDA inspection later is possible but is a separate, more expensive plan.

The numbers in brief (*illustrative; see Section 8*):

	Year 1	Year 2	Year 3
Head processed	200	300	375
Revenue ($\approx$ \$930/head)	\$186,000	\$279,000	\$348,750
Operating costs	\$112,071	\$152,185	\$163,114
Debt service (80% of \$471,576 at 8.5%, 10 yr)	\$56,130	\$56,130	\$56,130
Cash left for owner pay and reserves	\$17,799	\$70,685	\$129,507

- **Capital required:** about **\$470,000**, dominated by the mobile kill-and-cut unit ($\sim$ \$300,000 list).
- **Break-even with a \$50,000 owner draw:** about **275 head a year**, about 5–6 a week.
- **Capacity:** two people top out near **300–400 head a year**, because cutting takes about 8 labor-hours per custom beef.

The three things that decide go/no-go:

1. **Zoning for the cooler site.** In Frederick County’s Rural Areas (RA) district, any place where meat is “processed” is a *slaughterhouse* use, which needs a conditional use permit. A commercially zoned parcel with public sewer may be simpler.
2. **Wastewater.** Blood and wash water are regulated “industrial wastes”. Blood capture plus public sewer at the cooler is the clean route.
3. **Equipment cost and labor.** Real quotes for the mobile unit, and a second skilled cutter.

Financing approach. No grant program funds a new for-profit processor directly. The realistic stack is:

- **Farm Credit of the Virginias** equipment debt;
- **Virginia Small Business Financing Authority cash collateral** to cover lenders’ discount on specialized equipment;
- the **cooler as an AFID Infrastructure grant project** ($\leq$ \$50,000, applied for by a county);
- a free or low-fee **VAFAIRS** business-plan review.

Immediate next step: a 90-day diligence sprint (Section 11), starting with a written zoning use determination and equipment quotes.

2. The Business

2.1 Service

Step	Where	What happens
1. Booking	Phone / online	Owner books a date and signs the cut sheet. For shared animals, the list of owners is recorded before slaughter (VDACS requirement)
2. Harvest	Owner's farm	Mobile unit kills, skins and eviscerates. Offal is composted on that farm where allowed, or tubbed for the renderer. Blood is captured , not spread. Carcass chills in the unit's onboard cooler
3. Hang and age	Company cooler	Sides or quarters hang 7–21 days at about 34°F
4. Cut and pack	Mobile unit docked at the cooler	Cut to the owner's sheet, ground, vacuum-packed, labelled "Not for Sale", frozen
5. Pickup	Company cooler	Owner collects. Optional delivery

2.2 Customers

- **Primary:** cattle producers selling freezer beef by the half or quarter, who currently wait months for a slot or haul 30–100+ miles.
- **Secondary:** homesteaders and small herds finishing a few head for their own households.
- **Not customers:** retailers, restaurants, or anyone buying packaged meat. That requires inspection.

2.3 Value proposition

The service is priced at or above fixed-plant rates. It sells **time and convenience**, not a discount:

- no months-long queue
- no hauling live cattle, and no stress on the animals
- a harvest date the producer chooses, on their own farm
- one provider from field to freezer

2.4 Ownership structure (decision pending)

Structure	Opens	Costs
Sole operator LLC	Simplest; Farm Credit (as a farm-related service business), VSBFA	No co-op grants; FSA operating loans barred
Farmer-majority LLC or cooperative	Farm Credit eligibility on the cleanest basis; FSA loans for members buying co-op stock; USDA Value-Added Producer Grants	Governance overhead. Co-op governance problems are a documented cause of mobile-unit failure

3. Market Analysis

3.1 Demand signals

Signal	Value	Source / year
Regional USDA plants' utilization	125–150% of normal capacity	PEC / American Farmland Trust / Fuller Consulting, 2021
Producer waits	6–9 months before COVID; 12–18 months at peak	PEC 2021
Statewide waits	18–24 months	VDACS five-year plan (RD894), 2023
Nearest USDA plant	Books 3–5 months ahead	Blue Ridge Meats, Middletown, 2026
Neighbouring county	92% of surveyed farmers want another abattoir; Fauquier farmers travel to Stephens City and Middletown	Fauquier County survey (reported Nov 2024)
Binding constraint at existing plants	Shortage of trained meat cutters, at every regional plant	PEC 2021; Missouri peer-reviewed survey 2023

3.2 Cattle within reach (NASS 2022 Census of Agriculture, inventory Dec 31, 2022)

County	Cattle & calves	Farms selling direct to consumers
Frederick	10,720	9%
Clarke	10,686	12%
Warren	4,885	7%
Shenandoah	31,990	7%
Rockingham	87,565	8%
Augusta	86,078	7%
Total	231,924	

These counts include dairy. The immediate Winchester area (Frederick, Clarke, Warren) holds only about 26,000 head, and three-quarters of the Valley herd is 60–100 miles south. Basing the cooler in Shenandoah County, or near the I-81 corridor further south, widens the reachable herd *[inferred]*. At 300 head a year the business needs about **0.13% of the six-county inventory**.

3.3 Competition

Five Very Small federally inspected cattle plants serve Winchester to Harrisonburg (FSIS directory, 2026). Four of the five killed fewer than 1,000 head of all species in the trailing year.

Plant	Town	Known pricing
Gentle Harvest Custom Processing	Winchester	Not published
Blue Ridge Meats	Middletown	Not published; 3–5 month booking
Crabill’s	Toms Brook	Not published
Gore’s Processing	Edinburg	Not published
T&E Meats (Honest Meats)	Harrisonburg	\$75 kill + \$1.05/lb cut & pack (2025)
<i>Seven Hills (reference)</i>	<i>Lynchburg, ~175 mi</i>	<i>\$1.20–1.30/lb all-in, no lead time (2026)</i>

- **Potential partner or competitor:** the **ShenValley Meat Processing Cooperative** (Page County) received an \$87,450 GO Virginia planning grant in 2025 to address aging, storage and fabrication bottlenecks. It plans to outsource “live receiving”.
- **Unknown:** existing mobile custom operators in the Valley. The VDACS custom-permit roster will show them.

4. Regulatory Path

4.1 Why custom-exempt

	Custom-exempt (<i>this plan</i>)	Virginia state inspection	USDA inspection
Who may buy the meat	Only the owner of the live animal	Anyone, in Virginia only (Virginia is not in the CIS interstate program)	Anyone, any state
Permit	Free VDACS Custom Permit (2VAC5-210-30), renewed each June 30	Grant of inspection (form VDACS-03090); HACCP plan; 90-day validation	FSIS grant; HACCP; inspector at every kill
Mobile units	Expressly allowed, slaughter and processing	Not addressed	Same rules as a fixed plant, plus per-site paperwork
Product marking	“Not for Sale”	Virginia inspected mark	USDA mark

Rules the business must follow (9 CFR 303.1; VDACS custom guide):

- Records of every owner’s name and address.
- Sanitation under 9 CFR 416.1–416.6.
- Humane slaughter.
- Removal and segregation of specified risk materials (SRMs). For cattle 30 months or older: skull, brain, eyes, spinal cord and vertebral column. For all cattle: tonsils and distal ileum.
- **Charging only a service fee**, never a price for the meat.
- Shares are allowed if the owners are listed before slaughter.

Pending federal law (PRIME Act; the farm-bill custom-exempt pilot, capped at 5 facilities per state) could later allow limited in-state direct sales. **The plan assumes neither.**

4.2 Permits and approvals checklist

Item	Agency	Status
Custom Permit, covering the mobile kill-and-cut unit and the cooler	VDACS Office of Meat & Poultry Services	Ask whether the site is inspected and what checklist applies
Zoning use determination, then a conditional use permit or a site on a commercial parcel	Frederick County (or Shenandoah / Clarke / Page)	Critical path
Building permit, or the farm-building exemption (Va. Code § 36-97/99, on a working farm only)	County building official	Ask
Sewer connection and any pretreatment approval	Frederick Water / regional sewer authority; DEQ for pretreatment plans	Ask about discharge limits and surcharge
Private well (if no public water): coliform test every 6 months	Self / VDH	Below waterworks thresholds
Position on blood and wash water at farm kills	DEQ Valley Regional Office	Ask
Commercial driver's licence (if towing a semi-trailer)	DMV	—
Business registration, sales and use tax status on services	SCC / Virginia Tax	Confirm

4.3 Zoning: the main risk

In Frederick County's Rural Areas (RA) district:

- **Slaughterhouses are a conditional use**, not allowed by right. The county defines a slaughterhouse as an establishment “primarily engaged in the slaughtering **or processing** of meats”.
- The RA district has **no cold-storage use**.
- Storage of farm products is allowed by right only if the farm produced more than half of what it stores.

So a cooler where the mobile unit docks to cut is almost certainly a conditional “slaughterhouse” use. The permit would require:

- buildings at least 100 ft from property lines
- a 20,000 sq ft building cap
- **all operations under roof**, meaning a canopy or bay over the docked unit
- screening, plus public hearings

The alternative is a **commercially or industrially zoned parcel with public sewer**.

5. Operations

5.1 Equipment

Item	Option	Price (source, year)
Mobile kill-and-cut unit	Sentinel 53 ft semi: kill room, 10 ft cooler, 18 ft butcher room (saw, grinder, vac sealer), 35 kW generator, shore-power switch	From \$300,000 list (2026)
	Mobile Processing Trailers & Supply turnkey gooseneck, 38–44 ft (4–8 beef cooler)	Quote only. A gooseneck is likely friendlier on farm lanes <i>[inferred]</i>
	Alternative: kill-only trailer + a separate cut-room trailer parked at the cooler	Kill-only \$72k–\$115k (Sentinel 2026); cut-room trailer: quote
Tow vehicle	Semi tractor (Class A CDL), or a heavy pickup for a gooseneck	Placeholder \$60,000 used
Hanging cooler	About 400 sq ft for 20–30 beef, 34°F, rail and trolleys	CoolBot walk-ins from \$5,100 each (2026) up to plant-standard $\approx$ \$300/sq ft. Plan uses \$60,000 placeholder
Docking utilities	Shore power, potable water, roofed and bermed wash-down pad to sewer, blood and offal tubs	Included in placeholder

5.2 Capacity and labor

Task	Labor (NMPAN/Lorentz, 2010)
Harvest	$\approx$ 7 beef per person per 8-hour day
Custom cutting (owner’s cut sheet)	$\approx$ 8 labor-hours per beef
Standardized cutting	$\approx$ 4.5 labor-hours per beef

- **Typical week *[inferred]***: 2 harvest days and 3 cut-and-pack days at the cooler.
- **Ceiling with two people**: about **6–10 head a week**. The higher end needs a simplified cut menu.
- **Labor available in Year 3**: owner ($\approx$ 2,500 h) plus one full-time employee plus 1,000 seasonal hours $\approx$ 5,580 h.
- **Labor needed for 375 head**: $\approx$ 3,430 h of direct work, plus travel, clean-up and admin.
- **Wages**: Virginia butchers and meat cutters average **\$20.03/hr** (BLS, May 2025). The RappCE butchery course (\$900) teaches retail cutting; carcass breaking will need on-the-job training.

5.3 Waste

Stream	Plan	Rule
Offal, head, bones	Compost on the farm where the animal was killed, or tubbed for the renderer within 7 days	9VAC20-81-95 D.4 conditionally exempts carcass composting at the farm of generation; confirm DEQ's reading
SRMs from cattle ≥ 30 months	Segregated and dyed for the renderer; surcharge to the owner	21 CFR 589.2001 feed ban
Blood and first rinse	Captured and hauled with the offal	Captured blood cuts wastewater strength from 1,000–2,500 mg/l BOD to about 127 mg/l
Wash water at farm kills	Onboard holding tank <i>[inferred]</i>	DEQ “industrial waste”; whether the agricultural exclusion applies is unsettled
Wash water at the cooler dock	Public sewer , with a grease and solids interceptor	Frederick Water specifications; septic would need an engineered system
Renderer	Darling Ingredients (Winchester or Linville)	The only renderer statewide (RD894). Fees and minimums unknown

5.4 Scheduling and season

- **Booking:** committed slaughter dates booked ahead, with a deposit and a no-show fee. This is the documented fix for no-shows.
- **Peak:** demand peaks September–December. February–April is slow, so use it for maintenance, training and winter-finished cattle.
- **Cooler space:** cooler rail space is the scheduling constraint, at about 2–3 weeks of kill volume on the rail.

6. Pricing

Per-head price for a 750 lb hanging-weight steer. The fee shape follows operating mobile services; the levels come from regional benchmarks.

Line	Plan price	Benchmark (source, year)
Harvest (farm-call kill fee)	\$125/head	\$100–150 (Backyard Butchery, OK, 2025); \$75–125 (Beaverhead Meats, MT, 2026); \$75 at a fixed plant (T&E, 2025)
Mileage	≈ \$30/head average	\$2.65/mile one way (Backyard Butchery, 2025)
Daily minimum	\$500 per farm visit	Backyard Butchery, 2025
Cut, vac-pack, label	\$1.00/lb hanging (\$750)	\$0.90 (Backyard Butchery, 2025) – \$1.05 (T&E, 2025)
Extras (aging past 14 days, SRM handling for cattle ≥30 months, offal haul-off)	≈ \$25/head average	T&E: \$25/head/week aging, \$50 for cattle over 30 months; offal haul-off \$100 (Backyard Butchery)
Total	≈ \$930/head	Fixed USDA plant all-in ≈ \$860–975 (T&E / Seven Hills, 2025–26)

7. Start-up Capital

Item	Amount	Basis
Mobile kill-and-cut unit (Sentinel 53 ft list)	\$300,000	Sentinel 2026 list “from” price; equipment may be extra
Tow tractor, used	\$60,000	Placeholder — quote
Hanging cooler, ~400 sq ft + rail	\$60,000	Placeholder between CoolBot (~\$5k/unit) and \$300/sq ft
Sewer connection (2019 fees)	\$6,576	Frederick Water availability fee \$3,076 + lateral \$3,500 (2019)
Working capital & permits	\$45,000	Placeholder: CUP/engineering, permits, 2–3 months of operating cash
Total	\$471,576	

Financing assumption: 80% debt (\$377,261) at 8.5% over 10 years, with 20% owner or member equity (\$94,315). Annual debt service is \$56,130. Lender terms are a placeholder.

8. Financial Projections (illustrative)

	Year 1	Year 2	Year 3
Head processed	200	300	375
Revenue	\$186,000	\$279,000	\$348,750
Employee payroll incl. 15% burden	\$47,912	\$70,946	\$70,946
Packaging & supplies (\$50/head)	\$10,000	\$15,000	\$18,750
Renderer, SRM, hides (\$25/head)	\$5,000	\$7,500	\$9,375
Fuel & vehicle repair (\$120/harvest day)	\$6,000	\$9,000	\$11,250
Cooler utilities + sewer	\$8,130	\$8,295	\$8,419
Insurance (4.2% of revenue + workers comp code 2081)	\$13,030	\$19,444	\$22,374
Site lease	\$18,000	\$18,000	\$18,000
Accounting, legal, renewals	\$4,000	\$4,000	\$4,000
Operating costs	\$112,071	\$152,185	\$163,114
Operating cash flow (EBITDA)	\$73,929	\$126,815	\$185,636
Debt service	\$56,130	\$56,130	\$56,130
Cash for owner pay & reserves	\$17,799	\$70,685	\$129,507

Staffing assumed: owner (unpaid in these figures) + one full-time employee (2,080 h) in Year 1, and +1,000 seasonal hours from Year 2. Depreciation is about \$60,939/yr on 7-year straight-line. It is a tax item and is excluded from cash.

8.1 Break-even

Scenario	Head/yr to cover debt service	Head/yr to also pay the owner \$50,000
Base: \$300k kill-and-cut unit	210	274
Kill-only trailer + cut-room trailer (≈\$192k)	194	258
Unit at \$400k with options	226	289

8.2 Sensitivity to cutting price (Year 3 staffing, base capex)

Cut & pack price	Year 3 cash (375 head)	Head/yr to pay owner \$50k
\$0.90/lb	\$102,563	302
\$1.00/lb	\$129,507	274
\$1.10/lb	\$156,450	251
\$1.25/lb	\$196,866	223

What the numbers say *[inferred]*:

- **Year 1 is thin.** At 200 head the business barely covers debt service and leaves almost nothing for the owner. Plan the first year as a ramp, with savings or off-season income.
- **Viability comes in at roughly 275 head a year**, about 5–6 a week. That is inside the two-person ceiling but close to it.
- **Price matters more than equipment.** Each \$0.10/lb on cutting moves Year 3 cash by about \$27,000, while dropping the unit cost by \$108,000 moves it by about \$13,000 a year.
- **Not modelled:** land purchase, income taxes, a possible high-strength sewer surcharge, spoilage losses, and the owner's opportunity cost. Recalculate with real quotes before borrowing.

9. Funding Plan

Source	Use	Amount	Status (Oct 2026)
Farm Credit of the Virginias	Term loan or lease on the unit, tow vehicle and cooler. Eligible as a farm-related service business (12 CFR 613.3020), most cleanly if farmers own more than 50%	Bulk of the ≈\$377k debt	Confirm Frederick County is in its territory
VSBFA Cash Collateral (SSBCI 2.0)	Covers the lender's discount on specialized collateral	Up to 40% of the loan, max \$1M	Fees temporarily waived; the federal money is time-limited
VSBFA Capital Connect	Subordinate loan alongside the bank	Lesser of \$1M or 40% of project	Same
AFID Infrastructure grant	The cooler. Precedent: Piedmont Processing's cooler space for aging beef, 2022	≤ \$50,000, reimbursement, 1:1 match (2:1 if serving multiple small producers)	County must apply. Fall round closes Oct 30, 2026 ; target spring 2027
AFID Planning grant	County-sponsored feasibility study	\$20k (one county) / \$35k (several)	Rolling
VAFAIRS	Feasibility study and business-plan review	Free / low fee	Available
REAP (Year 2+)	Efficiency or solar for the fixed cooler	25% grant	Needs 12 months of operation
USDA VAPG	Working capital, <i>only if producer-owned</i>	≤ \$200k	Next round likely early 2027
FSA operating loans	—	—	Barred for a business that mainly processes others' animals

10. Risks and Mitigations

Risk	Likelihood / impact	Mitigation
Zoning denies or delays the cooler and dock	High / high	Get a written use determination first. Compare Shenandoah, Clarke and Page counties. Consider a commercial parcel with sewer
Wastewater: no sewer, or DEQ objects to farm wash water	Medium / high	Capture blood and first rinse, carry an onboard holding tank, choose a sewer site, get DEQ's position in writing
Can't hire or keep a second cutter	High / high	Owner cuts. Simplified cut menu. Seasonal cutter. RappCE pipeline. Pay above the \$20/hr mean
Volume below plan, or uneven across the year	Medium / high	Pre-sold harvest dates with deposits. Off-season pricing for winter-finished cattle. 100-mile service radius
Equipment cost above list, or long lead time	Medium / medium	Three quotes. Consider kill-only + cut-room trailers. Used units
Single renderer statewide	Medium / medium	On-farm composting where allowed. Confirm Darling's terms early
Food safety or customer illness	Low / high	9 CFR 416 sanitation SOPs even though not required for custom-exempt. Product liability cover
Share-sale structure challenged as meat sales	Low / high	Owners listed before slaughter. Service-fee-only invoices. Records kept under 303.1(b)(3)
Competition: ShenValley co-op or new mobile entrants	Medium / medium	Position as feeder or partner; the co-op plans to outsource live receiving
Spoilage (cooler failure)	Low / high	Alarm, generator backup, and the spoilage endorsement on the insurance policy

11. 90-Day Diligence Plan and Milestones

Weeks	Action	Decision it informs
1–2	Frederick County zoning use determination; same question to Shenandoah, Clarke, Page	Site strategy
1–3	Quotes: Mobile Processing Trailers & Supply turnkey gooseneck, Sentinel 53 ft, Friesla; kill-only + cut-room option	Capex
2–4	VDACS OMPS meeting: Custom Permit for the kill-and-cut unit and cooler; Valley custom-permit roster	Legal path; competitors
2–4	DEQ Valley Regional Office: composting exemption (D.4) and wash water. Frederick Water: sewer at candidate parcels	Waste plan
3–6	Survey 10–15 freezer-beef producers: head per year, wait, haul, willingness to pay	Volume and price
4–6	Darling Winchester: offal, blood, SRM, hides. Insurance agent quotes	Opex
6–8	Rebuild the pro forma with real figures. VAFAIRS review	Go / no-go
8–12	Lender meetings (Farm Credit, VSBFA). County AFID sponsorship request	Financing
After go	Site permits (CUP if needed), order the unit, build the cooler, hire. Target the first harvest season before the fall peak	Launch

Appendix A. Key Sources

All sources are archived in the mobile-beef-processing wiki (raw/), with URLs and fetch dates.

- VDACS OMPS, *A Guide to Red Meat Custom Exemptions; Applying for a Grant of Inspection* (Jan 2026); *Guide to Selling Meat and Poultry in Virginia* (May 2025)
- 9 CFR 303.1, 9 CFR 416, 9 CFR 310.22, 21 CFR 589.2001 (eCFR, 2026); 2VAC5-210; Va. Code Title 3.2 Ch. 54; Va. Code §§ 36-97, 36-99
- 9VAC20-81-95 and -310 (solid waste and composting); 9VAC25-32 (VPA); 12VAC5-590, -610, -613 (VDH)
- Frederick County Code Ch. 165 (Zoning), RA district and § 165-204.17 (Wayback snapshot, 2026-07-11)
- VDACS, *Five-Year Strategic Plan for Meat Processing Throughput Capacity* (RD894, Jan 2023)
- PEC / American Farmland Trust / Fuller Consulting regional processor study (2021); Matson / VAFAIRS, *Small-Volume Red Meat Processing in Virginia* (2020)
- FSIS MPI Directory (2026 snapshot); FSIS Cooperative Interstate Shipment and state-program lists

- NASS 2022 Census of Agriculture county profiles
- Sentinel Trailers 2026 price list; Mobile Processing Trailers & Supply; Friesla; CoolBot / Store It Cold (2026)
- NMPAN: MSU Manual (2010), product costing and throughput (Thiboumery, 2010), wastewater treatment, workers' compensation classifications, case studies (Island Grown, Puget Sound, Central Coast, Rusted Gate 2022)
- USDA ERS ERR-150 (2013); Gwin & Thiboumery, *JAFSCD* (2014); Syukron & Su, *JAFSCD* (2023)
- T&E Meats 2025 fee schedule; Seven Hills 2026 cut sheet; Backyard Butchery 2025 pricing; Blue Ridge Meats customer terms
- BLS OEWS May 2025 (SOC 51-3021); Iowa State PM 2077 small plant design guide (2009)
- VDACS AFID program guidelines; GO Virginia program manual (July 2026); VSBFA program pages; Farm Credit of the Virginias; 12 CFR 613; 7 CFR 761/764